

Justin / Ziheng Zhong 钟子恒

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EDUCATION

New York University — M.S. in Management & Analytics | GPA 4.0/4.0 New York, NY · 09/2025 – 05/2027

University of Toronto, St. George — B.Sc., Statistics & Mathematics (Double Major) Toronto, Canada · 09/2021 – 07/2025

TECHNICAL SKILLS

AI Coding & Agents: Claude Code, Codex; agentic workflow design, agent prompt engineering, multi-provider LLM routing, task-aware model selection, LLM output validation

Languages: Python, Java, TypeScript / JavaScript, SQL, Swift, R

Backend & Frameworks: Spring Boot, Node / Express, Cloudflare Workers (Hono, D1), REST APIs, Server-Sent Events (SSE)

Mobile & Tools: SwiftUI, SwiftData, WidgetKit; Git, Maven, Xcode, Oracle Data Modeler / SQL Developer

WORK EXPERIENCE

Xiaomi — Proretail App · Backend Developer Intern · Java 07/2024 – 08/2024

- Built three clue-statistics modules with drill-down detail, enabling comparative analysis of clue ownership across stores and employees.
- Reworked an existing clue filter and added two new filtering conditions plus a new interface for finer-grained queries.
- Cut total clue-count response time by ~**100 ms** after database reads through optimized SQL logic and elimination of unnecessary data transmission.

PROJECTS

Coin — iOS Spending Tracker + AI Backend · Swift, TypeScript, Cloudflare 05/2026 – Present

- Built and shipped a full-stack, local-first spending tracker (SwiftUI client + TypeScript/Hono backend on Cloudflare Workers), now in TestFlight; financial data stays **device-resident** for user-owned privacy.
- Exposed on-device expense data through a **local MCP server**: the model invokes typed tools (`query_expenses`, `aggregate_by_category`) via the MCP protocol to fetch only what it needs — only aggregated results, never raw transactions, reach the server for analysis — keeping data device-resident while enabling standardized tool-calling.
- Designed a **task-aware model-routing layer**: text from on-device OCR/ASR is analyzed by smaller models, while conversational weekly-yearly spending analysis is routed to larger models with deeper reasoning — matching model tier to each task's complexity and cost.
- Routed LLM calls across **Gemini and DeepSeek with server-side keys** (Sign in with Apple + App Attest hardware attestation, D1-backed sessions and metered usage); on-device Apple Intelligence serves as an offline fallback.

JobMuse — AI Career-Ops Workspace · React, Express, Node 05/2026 – Present

- Built a local-first workspace that **orchestrates a multi-stage job pipeline**: ingest job-board / JD links → auto-scrape candidate roles → LLM match-scoring → human-confirmed shortlist → per-role résumé tailoring.
- Implemented a **bounded concurrent worker pool** (shared-cursor queue + `Promise.all`, configurable concurrency, cooperative `AbortSignal` cancellation) to score multiple roles in parallel while capping load on AI providers.
- Backend owns provider keys and streams live job logs to the GUI over **SSE**, wired into application tracking, pipeline editing, and script runners.

NASA Space Apps Challenge · Team Leader, Backend · Java, JavaScript 09/2025 – 10/2025

- Designed a Java/Maven backend with **REST-style endpoints** fetching and serving data from multiple official NASA APIs, with robust handling of large-scale astronomical datasets and null/missing values.
- Collaborated on Monte Carlo impact simulations and orbital-dynamics calculations; defined API contracts powering the frontend's asteroid-trajectory and hazard visualizations.

Media Manager (Spotify API) · Team Leader · Java 09/2023 – 12/2023

- Tested Spotify API workflows in Java and built a token-retrieval shortcut; developed the GUI with Java Swing and implemented core add-song and search-song use cases.

LEADERSHIP

FRC — FIRST Robotics Competition · Team Leader, Team 8204 07/2017 – 06/2021

- Founded and captained Team 8204 in high school; previously robot driver and engineer for Team 5451, with top-four finishes at international competitions in Sydney, Shanghai, and Pittsburgh.